

Electrical Panel Inspection Report — MAC

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Panel Name:	MAC	Panel Code:	MAC
Location:	owner	Inspector:	admin
Inspection Mode:	switch_first	Duration / Frames:	11.7 s (0 frames)
Software Version:	0.1.0	Model Version:	cv-switch
Optical Sensor:	Direct Browser Camera (websocket-frame-stream)	Thermal Sensor:	simulator (Emissivity: Unspecified)
Calibration:	Simulated	Sensor Mode:	DEMO / SIMULATED

MISSING CONTEXT DISCLOSURE: Missing Electrical Load. Thermal readings could not be normalized for operational electrical loading or ambient drift.

Assessed Operational Risk Score: 60 / 100

❑ SIMULATED THERMAL DATA NOTICE: No radiometric physical thermal sensor was connected. All temperatures and thermal profiles in this document are synthetic test fixtures and MUST NOT be used for operational or safety decisions.

Visual & Thermal Capture Evidence

Visible Optical Feed



AI-Annotated Tracking Overlay



Radiometric / Thermal Image



1. Observed Physical Indicators (Factual Measurements)

Direct sensor observations prior to diagnostic inference or algorithmic interpretation.

#	Component	Type	Spot Tmax	Avg Temp	ΔT (Ambient)	Visual Condition
1	Wall Switch	wall_switch	58.1 °C	58.1 °C	+36.1 °C	Normal Appearance

2. Inferred Diagnostic Hypotheses (AI Diagnostic Interpretation)

Algorithmic fault hypotheses derived from thermal gradients, peer deviations, and visual evidence.

#	Inferred Fault	Severity	Confidence	Model / Rule Version	Diagnostic Description
1	Thermal Anomaly	HIGH	90.0%	cv-switch / rules-v1.4	Abnormal thermal pattern detected near the electrical switch

3. Incident Lifecycle & Reviewer Actions

No persistent incidents tracked for this inspection session.

4. Unresolved Observational Limitations

- Thermal sensor resolution cannot resolve sub-terminal wire contact interfaces.
- Operating load context was unrecorded during capture; thermal margin is uncompensated for load variations.
- Inspection utilized synthetic/simulated thermal data; measurements have no physical validity.

5. Recommended Qualified Follow-up

- Qualified electrician to perform physical torque verification on flagged connections using calibrated tools.
- Re-inspect during peak facility electrical demand to assess loaded thermal margins.

NON-CERTIFICATION NOTICE: This automated inspection report was generated by ThermoGuard AI as an inspection-assistance tool. This document does NOT certify electrical equipment safety, operational fitness, or compliance with NFPA 70E, IEEE, or OSHA standards. A physical examination by a licensed and certified electrical professional is required prior to taking equipment into or out of service.



Inspector Physical Sign-Off (Manual Acknowledgment)

Signature: _____ Date: _____

Printed Name: _____ License / ID: _____

NOTICE: This signature line is a manual physical sign-off placeholder for printed copies and does NOT constitute a cryptographic digital signature.